

Update On Multi-background Anomaly Detection

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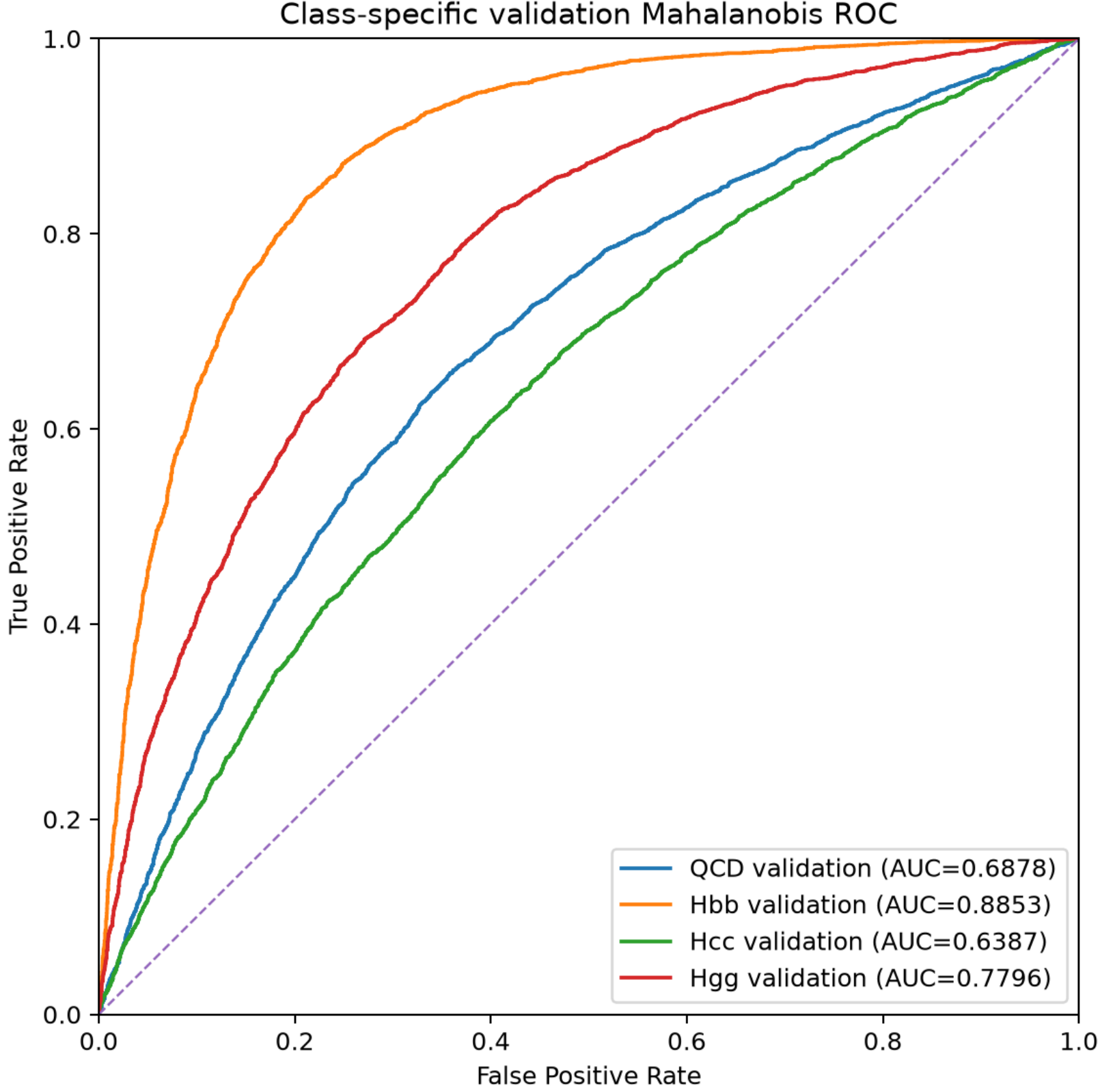
Jul 20, 2026

Ablation Test

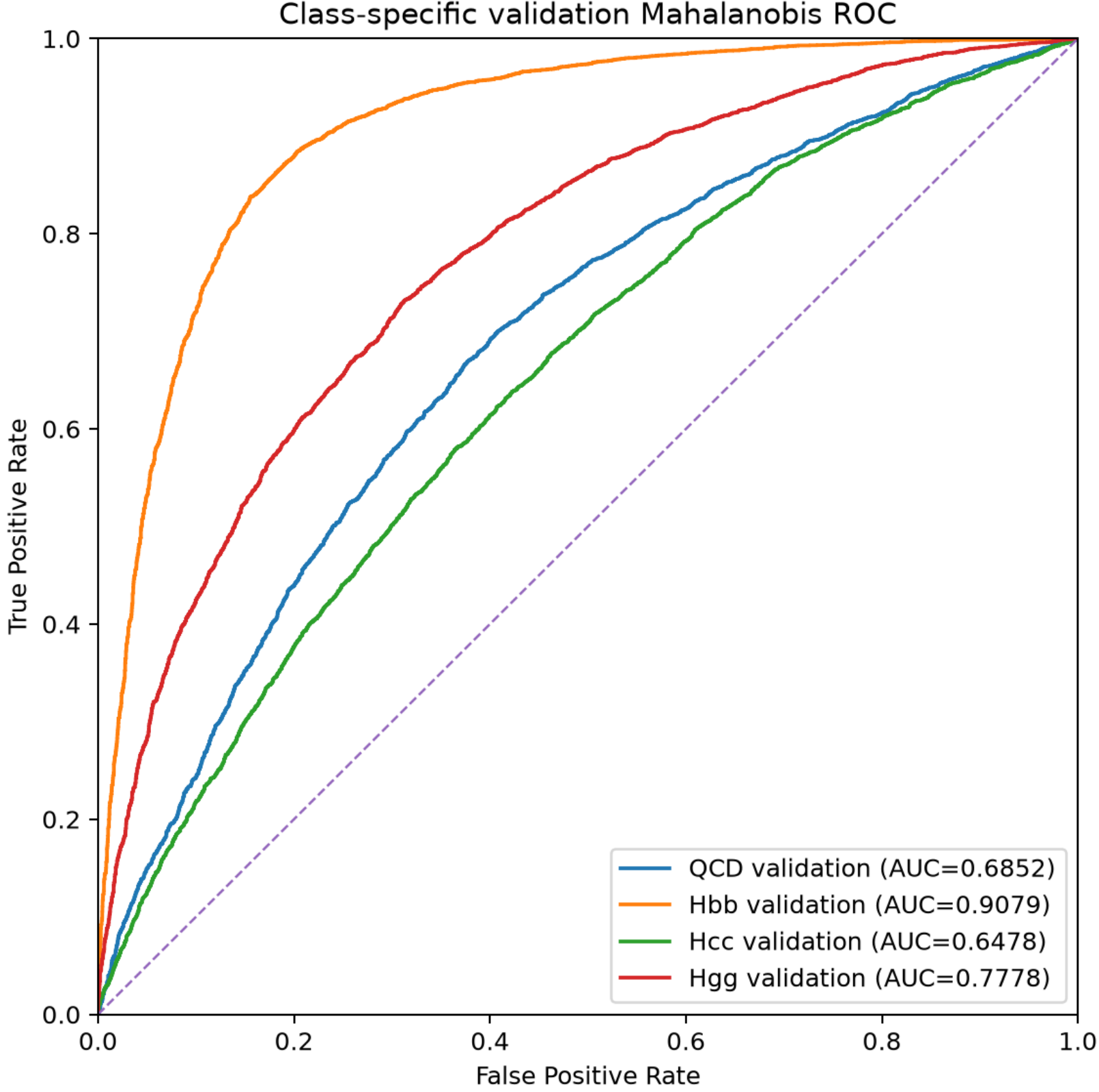
- Train models with different optimization targets and compare performance differences.
 - LeJEPA, LeJEPA + classification, LeJEPA + triplet, LeJEPA + triplet + classification (unfinished)
- Background: QCD + Hbb + Hcc + Hgg
- Signal: Wqq
- Triplet margin: 0.2
- Epochs: 12
- Steps per epoch: 4000
- Hidden & repr dim: 32
- Num layers: 4

Ablation Test

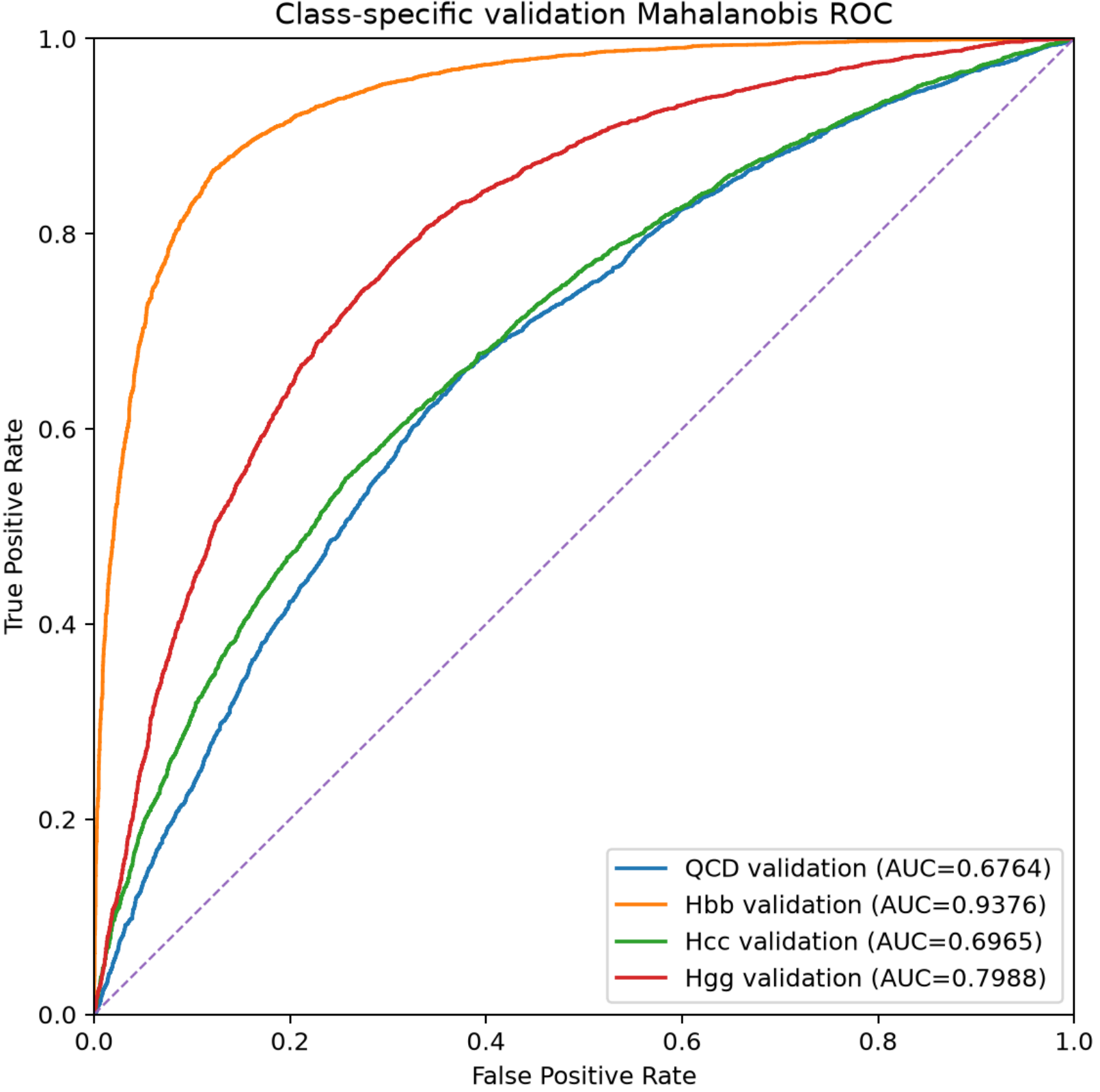
LeJEPA



LeJEPA
+ triplet

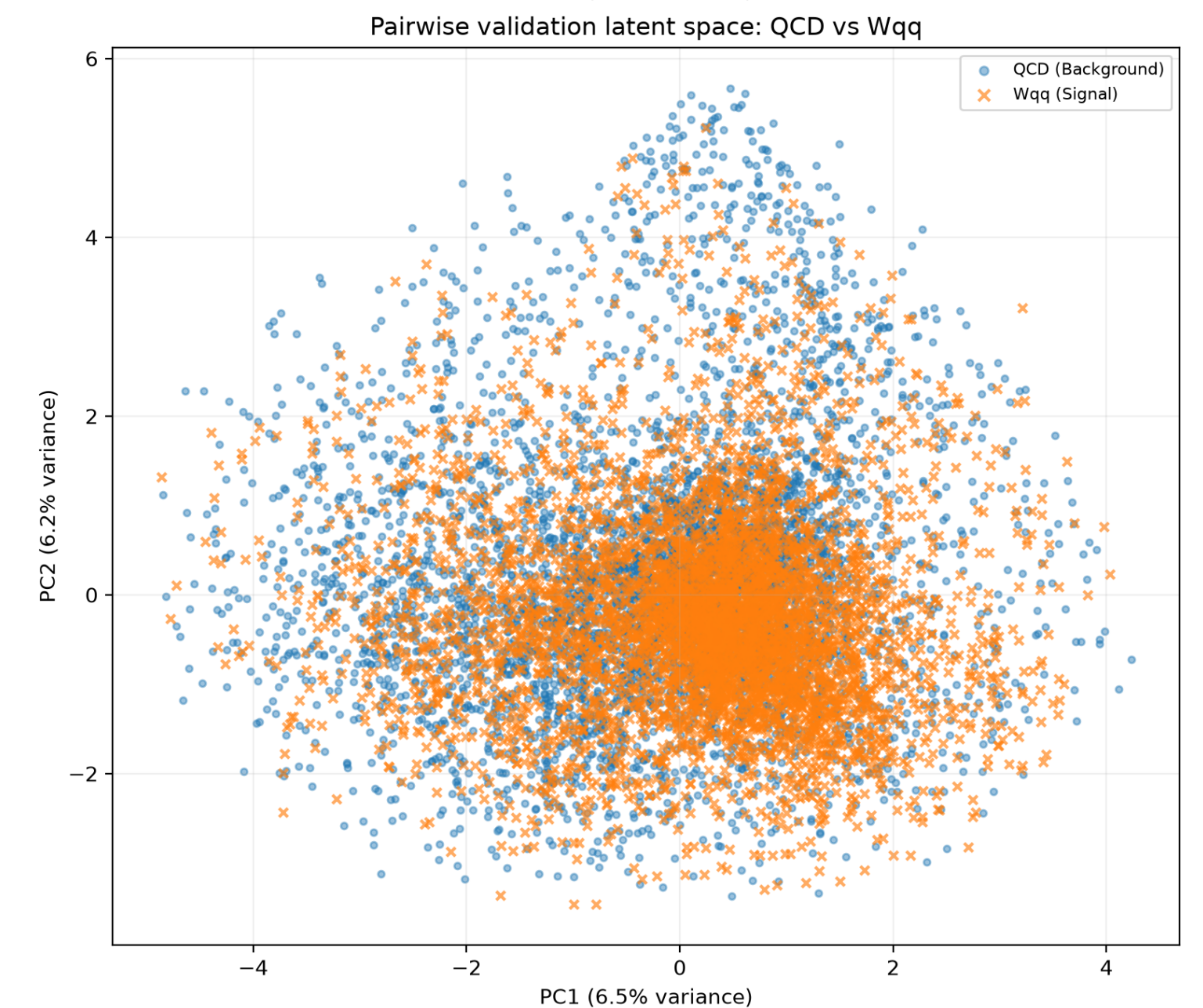
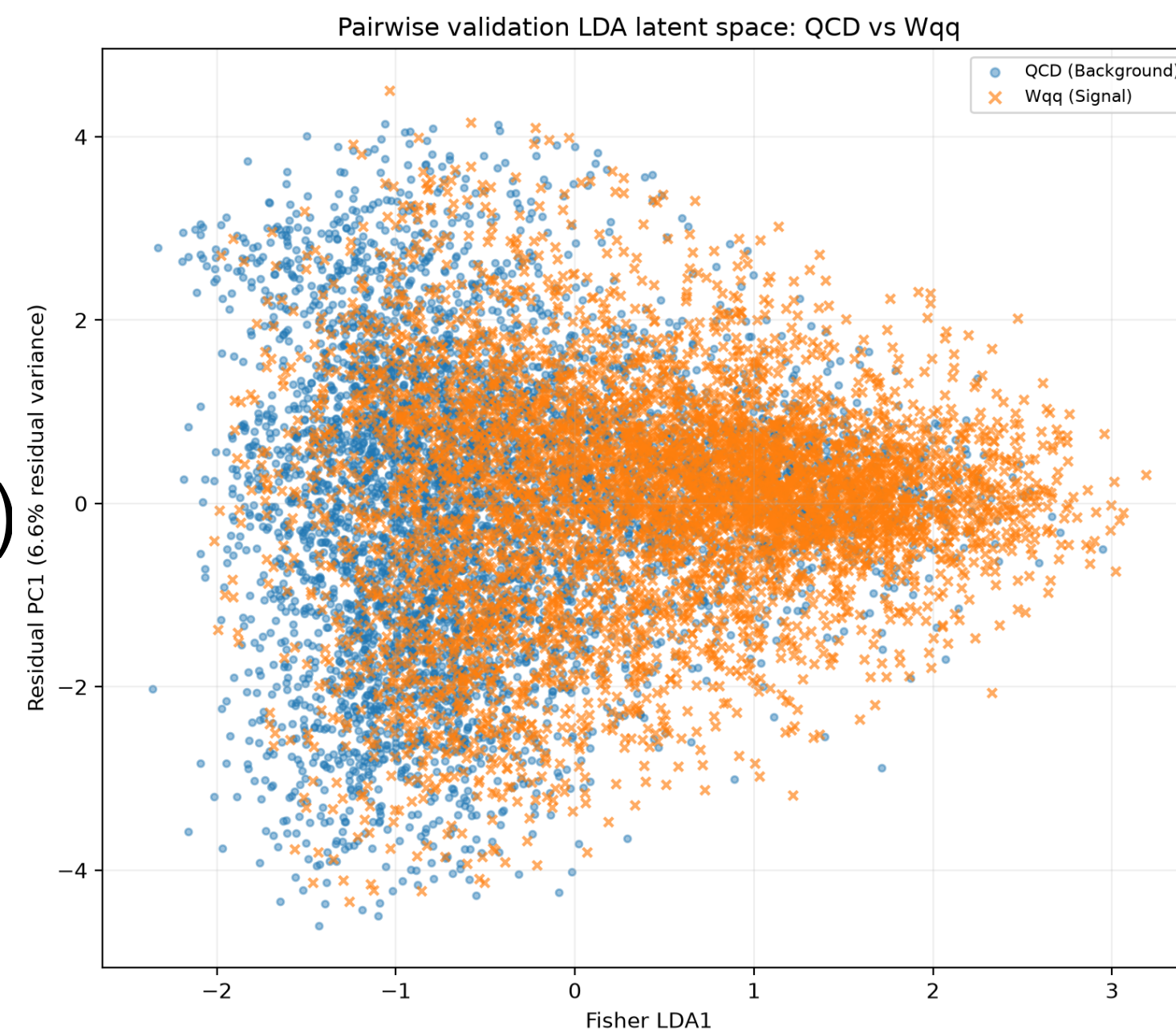
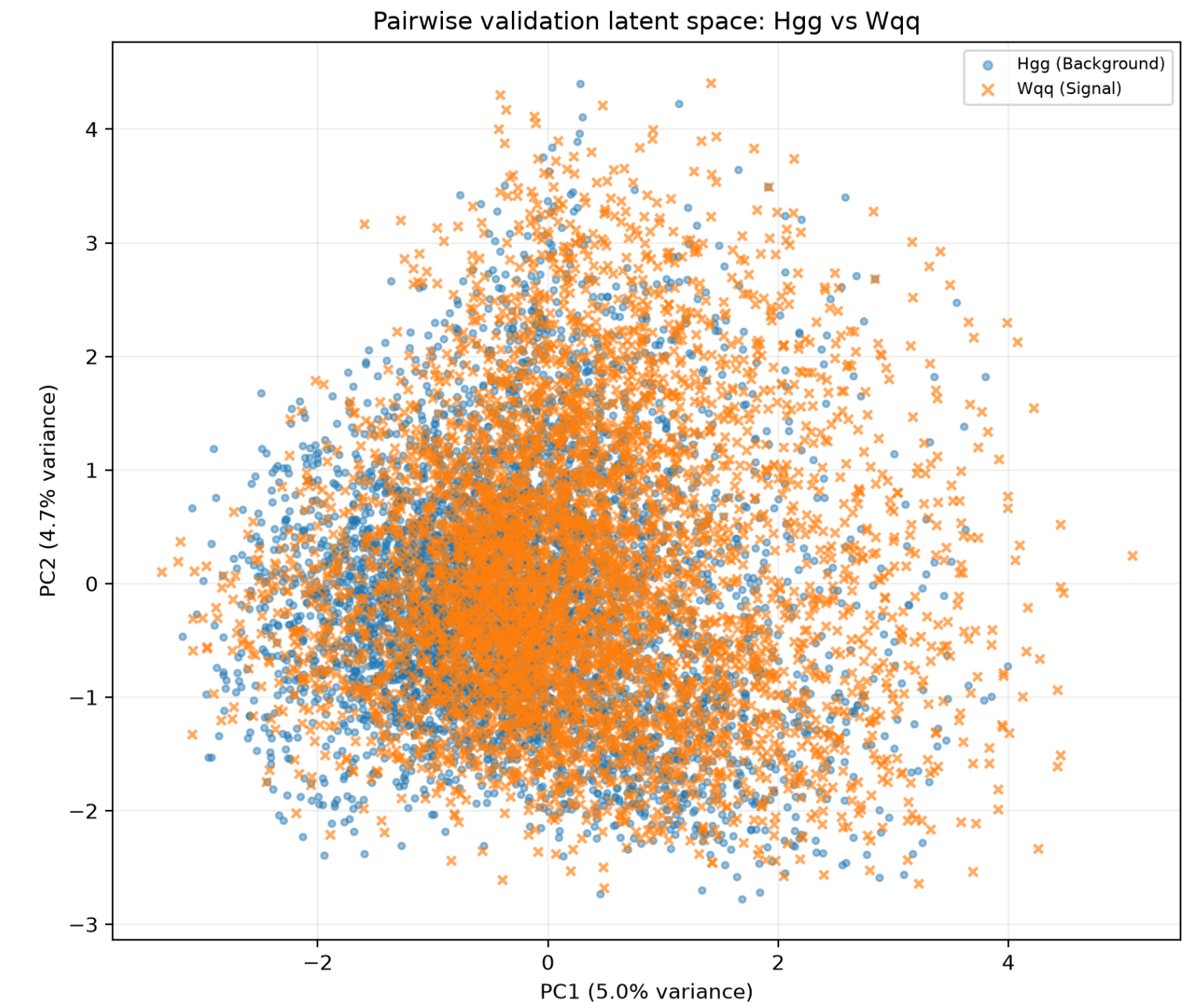
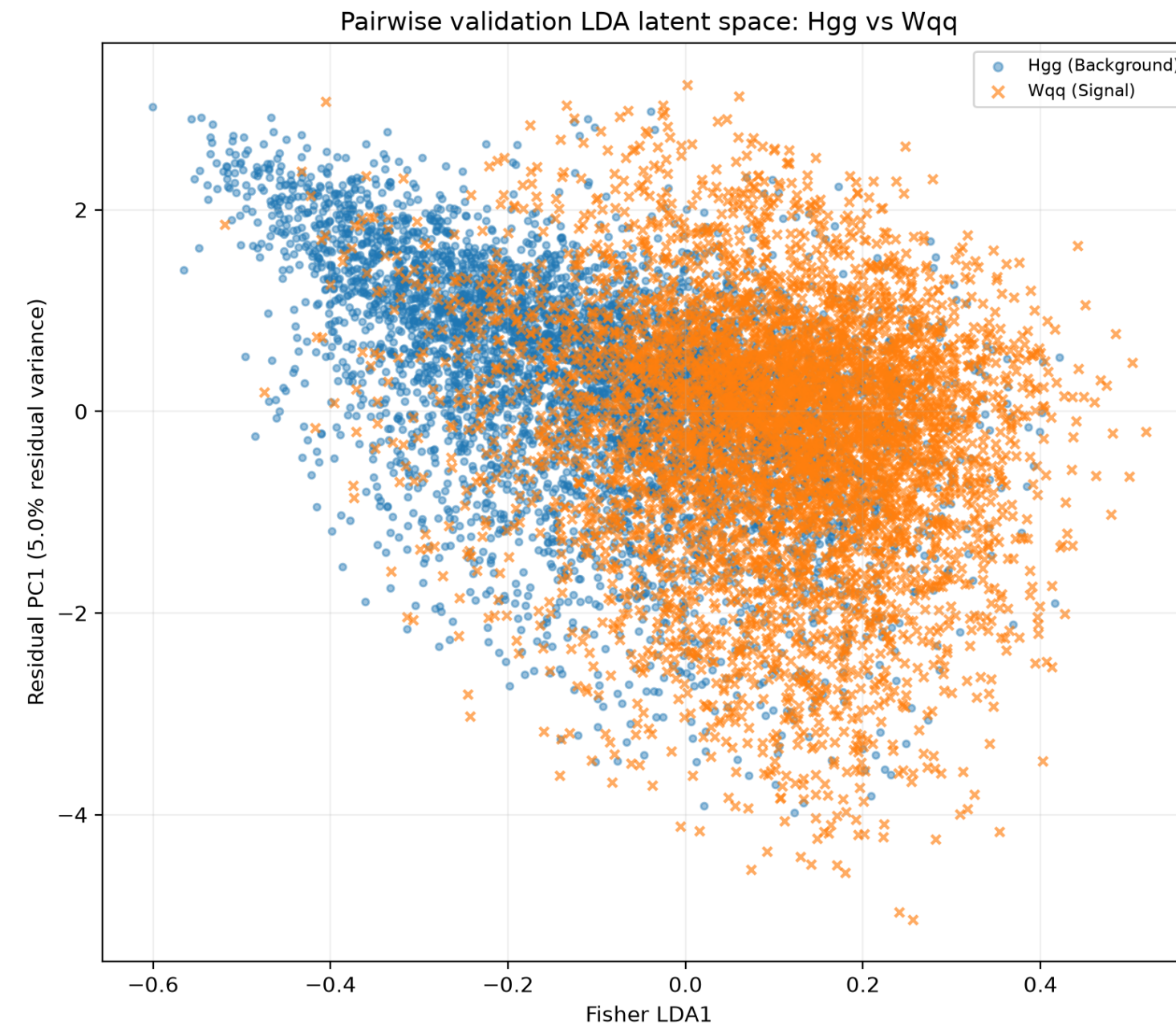


LeJEPA
+ classification



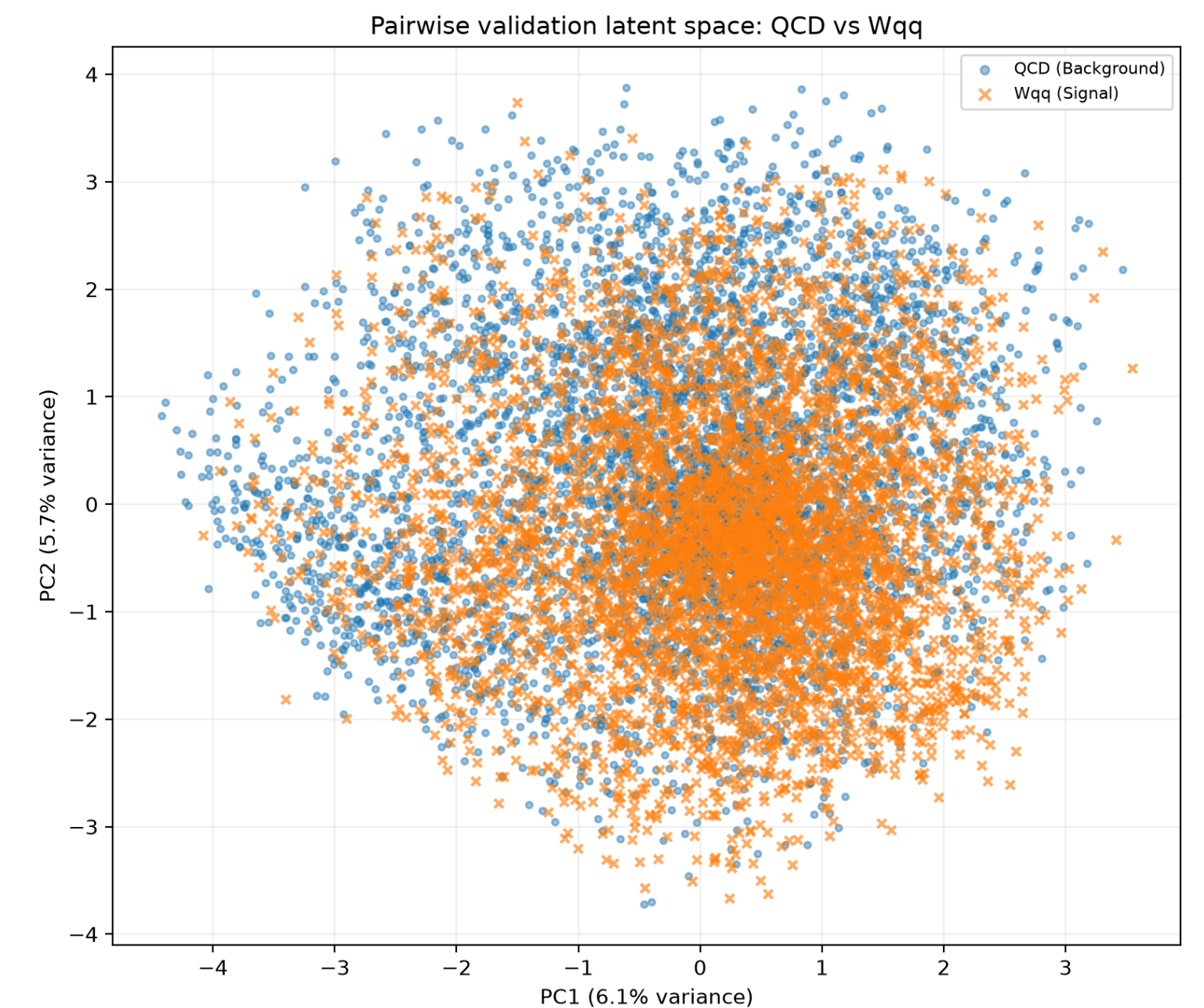
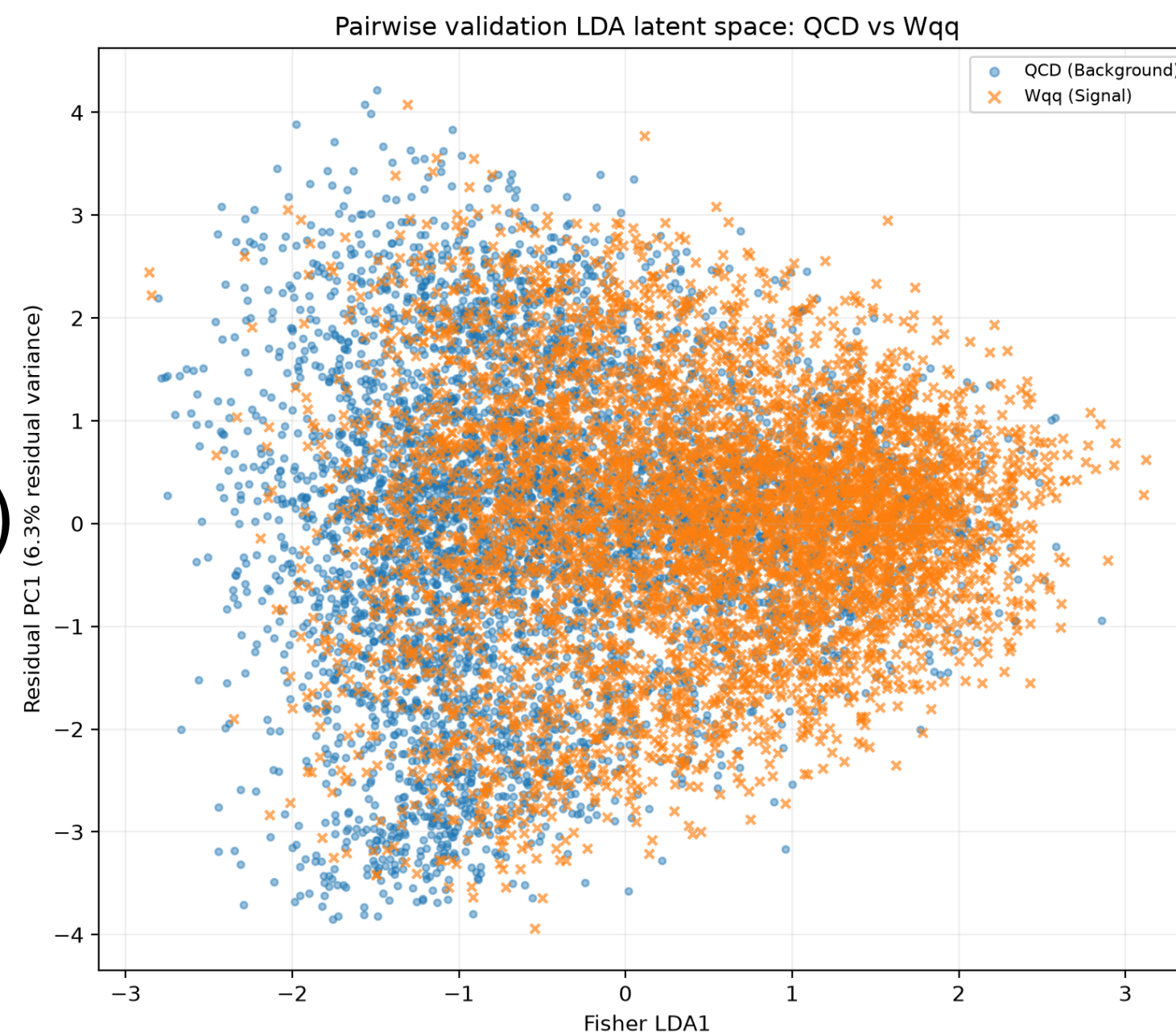
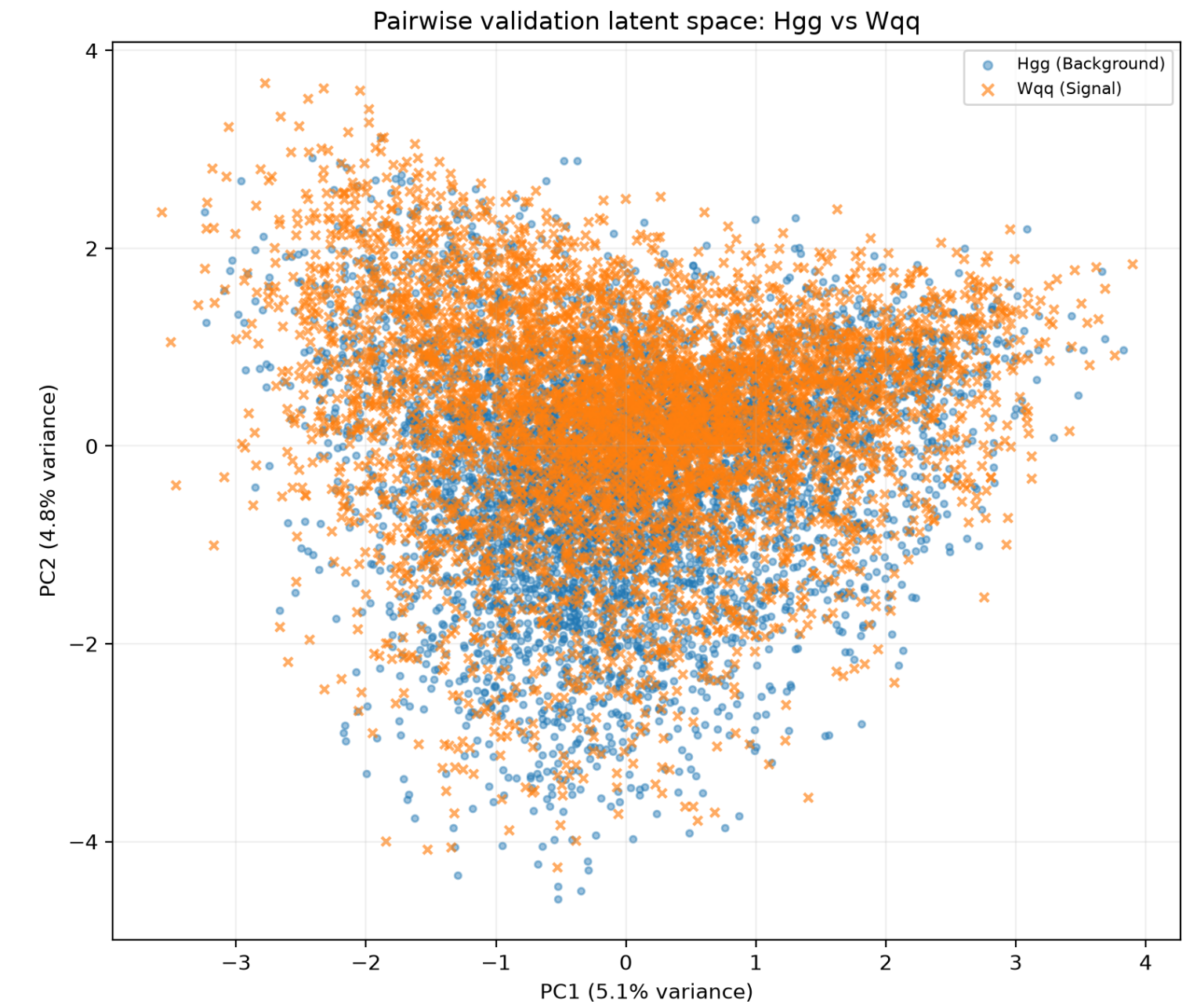
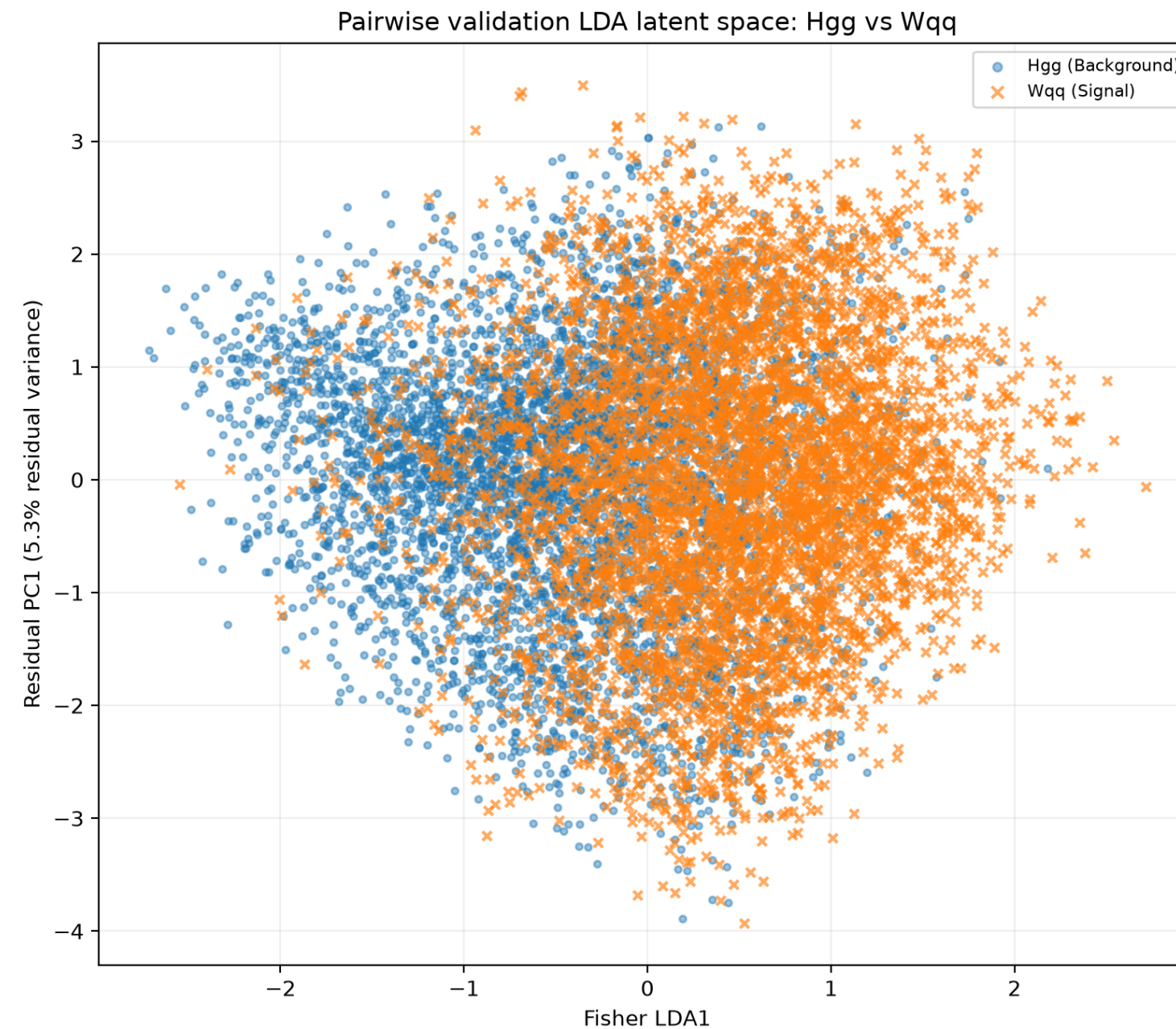
LDA Analysis

- LeJEPA
- Left: LDA + residual PCA
- Right: PCA
- Top: Hgg vs Wqq (AUC 0.8853)
- Bottom: QCD vs Wqq (AUC 0.6878)



LDA Analysis

- LeJEPA + triplet
- Left: LDA + residual PCA
- Right: PCA
- Top: Hgg vs Wqq (AUC 0.9079)
- Bottom: QCD vs Wqq (AUC 0.6852)



LDA Analysis

- LeJEPA + classification
- Left: LDA + residual PCA
- Right: PCA
- Top: Hgg vs Wqq (AUC 0.9376)
- Bottom: QCD vs Wqq (AUC 0.6764)

